

HANSHIKA AWASTHI

Kanpur, Uttar Pradesh | +91-9369811689 | hansikaawasthi11256@gmail.com | [LinkedIn](#) | [LeetCode](#)

PROFILE

Computer Science undergraduate with research experience in machine learning and quantitative modeling, and hands-on experience shipping full-stack AI applications (LLM-powered chatbots, voice assistants, BI dashboards) for real clients.

EDUCATION

Allenhouse Institute of Technology, Kanpur, UP 2024 – 2028

B.Tech in Computer Science and Engineering

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Discrete Mathematics, Computer Networks

RESEARCH & INDUSTRY EXPERIENCE

Research Intern – Indian Institute of Science (IISc), Bengaluru 2025

Supervised by Prof. Ratikanta Behera

Implemented a regime-switching model to forecast gold and silver prices, applying Gaussian Mixture Models and Monte Carlo simulation to capture volatility regime shifts in commodity markets.

Collaborated with faculty on methodology design and result validation for an ongoing research paper targeting Q1 journal submission.

Freelance AI/ Full-Stack Developer – Independent Client Work 2025 – Present

Built and deployed a full-stack AI shopping assistant chatbot for an Indian D2C jewellery brand: Flask + Gemini API backend with rate limiting and a server-side system prompt, plus a floating chat-bubble frontend with in-chat product recommendation cards and marketplace redirect logic.

Designed a Power BI dashboard from scratch for project completion tracking, building custom DAX measures and gauge visualizations across Modeling, Drawing, and QC status.

PROJECTS

Browser-Based Voice Assistant | Flask, Gemini API, Web Speech API 2025

Built a voice-driven conversational assistant using the Web Speech API for input/output, with a Flask + Gemini backend for natural language understanding, published as a portfolio project on GitHub.

Hybrid Movie Recommendation System | Python, TF-IDF, SVD, Pandas 2025

Built a recommendation engine combining content-based and collaborative filtering to generate personalized movie suggestions.

Evaluated using Precision@K and RMSE metrics on the MovieLens dataset.

Gold & Silver Price Predictor | Python, LSTM, Scikit-learn, Matplotlib 2025

Developed a time-series forecasting model to predict precious metal prices using historical market data.

Compared LSTM, Linear Regression, and Random Forest models using MSE and MAE benchmarks.

RESEARCH & PUBLICATIONS

Bridging Content and Collaborative Filtering: A Hybrid Recommendation Framework Using TF-IDF and SVD July 2025

Accepted at the 3rd International Conference on Machine Learning, Cryptology & Network Security (ICMLCS 2025), 26–27 July 2025.

Proposed a hybrid recommendation system combining TF-IDF (content-based) and SVD (collaborative filtering) to improve accuracy and address the cold-start problem.

Metal Price Forecasting Using Machine Learning 2025 – Present

Ongoing research studying gold and silver price movement patterns across market regimes using supervised ML models, regression, time-series forecasting (LSTM), and feature engineering on historical commodity datasets.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, HTML, CSS, JavaScript

ML & Data: NumPy, Pandas, Scikit-learn, Matplotlib, NLTK, Time-Series Forecasting (LSTM), Regime-Switching Models, Monte Carlo Simulation

Applied AI: LLM API Integration (Gemini), Flask, Conversational/Voice Interfaces (Web Speech API)

BI & Databases: Power BI, DAX, MySQL, SQLite

Tools: Git & GitHub, VS Code, Jupyter Notebook, Google Colab, LaTeX, MS Excel

ACHIEVEMENTS & CERTIFICATIONS

Certificate of Excellence in Research July 2025

Awarded by Allenhouse Institute of Technology for presenting "Bridging Content and Collaborative Filtering: A Hybrid Recommendation Framework Using TF-IDF and SVD" at ICMLCS 2025.

Certificate of Participation – Multimedia Presentation Competition November 2024

AllenTech IT Club, Allenhouse Institute of Technology – presented on phishing attack vectors, detection, and mitigation.